

B.E. I EXAMINATION JAN' 2024
Electronics & Instrumentation Engg.
2AMRC1: Applied Mathematics-II

Time: 3 Hrs.]

[Max. Marks: 60]

Note: Attempt any TWO parts from each Question. Each carries equal marks.

- Q.1. (a) Find the normal form of the matrix A and hence find its rank, where (06)

$$A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

- (b) Test for consistency and solve (06)

$$5x + 3y + 7z = 4; \quad 3x + 26y + 2z = 9; \quad 7x + 2y + 10z = 5$$

- (c) Verify Cayley-Hamilton theorem for the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ and hence, find the (06)

matrix represented by $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$.

- Q.2. (a) Solve the following equation : $(y^3 - 2x^2y)dx + (2xy^2 - x^3)dy = 0$ (06)

- (b) Solve : $(D^2 - 4D + 4)y = 8x^2 e^{2x} \sin 2x$ (06)

- (c) Solve the simultaneous differential equations : $dx/dt + y = \sin t$, $dy/dt + x = \cos t$ (06)

- Q.3. (a) Solve the following partial differential equations : (06)

(1) $p + 3q = 5z + \tan(y - 3x)$

(2) $\left[\frac{(b-c)}{a} \right] yz p + \left[\frac{(c-a)}{b} \right] zx q = \left[\frac{(a-b)}{c} \right] xy$

- (b) Solve : $(D^2 + DD' - 6D'^2)z = \cos(2x + y)$ (06)

- (c) Solve $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$ by the method of separation of variables where $u(x, 0) = 6e^{-3x}$. (06)

- Q.4. (a) A and B take turns in throwing dice, the first to throw 10 being awarded the prize. Show that if A has the first throw, their chance of winning are in the ratio 12:11. (06)

- (b) Fit a second degree parabola to the given data by least squares method: (06)

x	1	2	3	4	5	6	7	8	9
y	2	6	7	8	10	11	11	10	9

- (c) Calculate the coefficient of correlation between the marks obtained by 8 students in mathematics and statistics: (06)

Students	A	B	C	D	E	F	G	H
Mathematics	25	30	32	35	37	40	42	45
Statistics	08	10	15	17	20	23	24	25

- Q.5. (a) Calculate the value of following for the cubic equation $x^3 + px^2 + qx + r = 0$, where the roots are α, β, γ : (06)

(1) $\sum \alpha^2 \beta^2$, (2) $\sum \alpha^2 \beta$, (3) $\sum \alpha^2$, (4) $\sum \alpha^3 \beta$

- (b) Solve by Cardan's method : $x^3 + x^2 - 16x + 20 = 0$ (06)

- (c) (1) Write a short note on fuzzy set. (2 marks) (06)
 (2) Explain the properties of fuzzy set with examples. (4 marks)
